SQL Query Implementation for E-Commerce Database (EcoCart)

Aim:

To retrieve, filter, sort, and analyse customer, product, order, payment, review, and inventory information in the EcoCart e-commerce database using SQL queries.

Objectives:

- To retrieve records from database tables using the SELECT statement.
- To filter records using WHERE conditions.
- To sort records using ORDER BY.
- To retrieve unique values using DISTINCT.
- To search products based on price, category, name, and stock availability.
- To combine related information from multiple tables.
- To generate business reports related to customers, orders, inventory, and product performance.

SQL Queries and Implementation

1.Basic SQL Queries:

1. Display all customer details:

```
SQL> SELECT * FROM customer;

CUSTOMER_I CUSTOMER_NAME       EMAIL                          PHONE ADDRESS             REGISTRAT
---------- ------------------- ------------------------------ ----------- ------------------- ---------
CUS001     Sanjay Kumar        sanjay@gmail.com               9876500001 Chennai             05-JAN-26
CUS002     Harini Devi         harini@gmail.com               9007654321 Coimbatore          08-JAN-26
CUS003     Praveen Raj         praveen@gmail.com              9876500003 Madurai             12-JAN-26
CUS004     Keerthana S         keerthana@gmail.com            9876500004 Salem               15-JAN-26
CUS005     Dinesh Kumar        dinesh@gmail.com               9876500005 Trichy              20-JAN-26
CUS006     Pavithra M          pavithra@gmail.com             9876500006 Erode               02-FEB-26
CUS007     Ashwin Anand        ashwin@gmail.com               9876500007 Tiruppur            10-FEB-26
CUS008     Nithya R            nithya@gmail.com               9876500008 Pondicherry         15-FEB-26
CUS009     Surya Prakash       surya@gmail.com                9876500009 Vellore             20-FEB-26
CUS010     Lavanya M           lavanya@gmail.com              9876500010 Thanjavur           25-FEB-26

10 rows selected.
```

2. Display all available products:

```
SQL> SELECT * FROM product;

PRODUCT_ID PRODUCT_NAME               CATEGOR      PRICE STOCK_QUANTITY
---------- ------------------------- ------- ---------- --------------
PRO001     Organic Oats              CAT001      300.00             45
PRO002     Natural Honey             CAT001      320.00             35
PRO003     Aloe Face Cream           CAT002      250.00             40
PRO004     Bamboo Comb               CAT003      100.00             90
PRO005     Bamboo Tray               CAT003      400.00             30
PRO006     Steel Lunch Box           CAT004      600.00             55
PRO007     Eco Diary                 CAT005      140.00             70
PRO008     Cotton Tote Bag           CAT006      180.00             65
PRO009     Eco Cotton Kurta          CAT008      850.00             25
PRO010     Organic Seed Pack         CAT009      110.00             95

10 rows selected.
```

3. Retrieve product names and prices:

```
SQL> SELECT product_name, price FROM product;

PRODUCT_NAME                    PRICE
------------------------- ------------
Organic Oats                    300.00
Natural Honey                   320.00
Aloe Face Cream                 250.00
Bamboo Comb                     100.00
Bamboo Tray                     400.00
Steel Lunch Box                 600.00
Eco Diary                       140.00
Cotton Tote Bag                 180.00
Eco Cotton Kurta                850.00
Organic Seed Pack               110.00

10 rows selected.
```

4. Display all orders:

```
SQL> SELECT c.customer_name,o.order_id,o.total_amount,o.order_status FROM customer c JOIN orders o ON c.customer_id=o.customer_id;

CUSTOMER_NAME       ORDER_ID   TOTAL_AMOUNT ORDER_STATUS
------------------- ---------- ------------ ------------------
Sanjay Kumar        ORD001           520.00 Delivered
Harini Devi         ORD002           250.00 Delivered
Praveen Raj         ORD003           600.00 Shipped
Keerthana S         ORD004           400.00 Delivered
Pavithra M          ORD006           850.00 Shipped
Ashwin Anand        ORD007           420.00 Delivered
Lavanya M           ORD010           520.00 Shipped
Surya Prakash       ORD009           200.00 Pending
Nithya R            ORD008           360.00 Pending
Dinesh Kumar        ORD005           180.00 Pending

10 rows selected.
```

5. Retrieve payment details:

```
SQL> SELECT * FROM payment;

PAYMENT ORDER_ID   PAYMENT_D PAYMENT_MODE    PAYMENT_STATUS  TRANSACTION_AMOUNT
------- ---------- --------- --------------- --------------- ------------------
PAY001  ORD001     01-SEP-26 UPI             Success                        520
PAY002  ORD002     02-SEP-26 Card            Success                        250
PAY003  ORD003     03-SEP-26 Net Banking     Success                        600
PAY004  ORD004     04-SEP-26 UPI             Success                        400
PAY005  ORD005     05-SEP-26 Cash            Pending                        180
PAY006  ORD006     06-SEP-26 Card            Success                        850
PAY007  ORD007     07-SEP-26 UPI             Success                        420
PAY008  ORD008     08-SEP-26 Card            Pending                        360
PAY009  ORD009     09-SEP-26 Net Banking     Failed                         200
PAY010  ORD010     10-SEP-26 UPI             Success                        320

10 rows selected.
```

2. Filtering Conditions Using WHERE:

1. Find products with a price greater than ₹5,000:

```
SQL> SELECT * FROM product WHERE price > 5000;

no rows selected
```

2. Display products available in stock:

```
SQL> SELECT * FROM product WHERE stock_quantity > 0;

PRODUCT_ID PRODUCT_NAME              CATEGOR        PRICE STOCK_QUANTITY
---------- ------------------------- ------- ---------- --------------
PRO001     Organic Oats              CAT001        300.00             45
PRO002     Natural Honey             CAT001        320.00             35
PRO003     Aloe Face Cream           CAT002        250.00             40
PRO004     Bamboo Comb               CAT003        100.00             90
PRO005     Bamboo Tray               CAT003        400.00             30
PRO006     Steel Lunch Box           CAT004        600.00             55
PRO007     Eco Diary                 CAT005        140.00             70
PRO008     Cotton Tote Bag           CAT006        180.00             65
PRO009     Eco Cotton Kurta          CAT008        850.00             25
PRO010     Organic Seed Pack         CAT009        110.00             95

10 rows selected.
```

3. Display completed orders:

```
SQL> SELECT * FROM orders WHERE order_status = 'Delivered';

ORDER_ID   CUSTOMER_I ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS
---------- ---------- ------------ ------------ --------------------
ORD001     CUS001     01-SEP-26          520.00 Delivered
ORD002     CUS002     02-SEP-26          250.00 Delivered
ORD004     CUS004     04-SEP-26          400.00 Delivered
ORD007     CUS007     07-SEP-26          420.00 Delivered
```

4. Find products with a rating above 4:

```
SQL> SELECT p.product_id, p.product_name, r.rating FROM product p JOIN review r ON p.product_id = r.product_id WHERE r.rating > 4;

PRODUCT_ID PRODUCT_NAME               RATING
---------- ------------------------- ----------
PRO001     Organic Oats                       5
PRO003     Aloe Face Cream                    5
PRO005     Bamboo Tray                        5
PRO007     Eco Diary                          5
PRO009     Eco Cotton Kurta                   5
```

3.Sorting Data Using ORDER BY:

1. Display products from lowest to highest price:

```
SQL> SELECT product_name, price FROM product ORDER BY price;

PRODUCT_NAME                      PRICE
------------------------- ------------
Bamboo Comb                     100.00
Organic Seed Pack               110.00
Eco Diary                       140.00
Cotton Tote Bag                 180.00
Aloe Face Cream                 250.00
Organic Oats                    300.00
Natural Honey                   320.00
Bamboo Tray                     400.00
Steel Lunch Box                 600.00
Eco Cotton Kurta                850.00

10 rows selected.
```

2. Display customers alphabetically:

```
SQL> SELECT * FROM customer ORDER BY customer_name;

CUSTOMER_I CUSTOMER_NAME       EMAIL                          PHONE ADDRESS             REGISTRAT
---------- ------------------- ------------------------------ ----------- ------------------- ---------
CUS007     Ashwin Anand        ashwin@gmail.com               9876500007 Tiruppur            10-FEB-26
CUS005     Dinesh Kumar        dinesh@gmail.com               9876500005 Trichy              20-JAN-26
CUS002     Harini Devi         harini@gmail.com               9007654321 Coimbatore          08-JAN-26
CUS004     Keerthana S         keerthana@gmail.com            9876500004 Salem               15-JAN-26
CUS010     Lavanya M           lavanya@gmail.com              9876500010 Thanjavur           25-FEB-26
CUS008     Nithya R            nithya@gmail.com               9876500008 Pondicherry         15-FEB-26
CUS006     Pavithra M          pavithra@gmail.com             9876500006 Erode               02-FEB-26
CUS003     Praveen Raj         praveen@gmail.com              9876500003 Madurai             12-JAN-26
CUS001     Sanjay Kumar        sanjay@gmail.com               9876500001 Chennai             05-JAN-26
CUS009     Surya Prakash       surya@gmail.com                9876500009 Vellore             20-FEB-26

10 rows selected.
```

3. Find the most expensive products first:

```
SQL> SELECT product_name, price FROM product ORDER BY price DESC;

PRODUCT_NAME                    PRICE
------------------------- ------------
Eco Cotton Kurta                850.00
Steel Lunch Box                 600.00
Bamboo Tray                     400.00
Natural Honey                   320.00
Organic Oats                    300.00
```

4. Display latest orders first:

```
SQL> SELECT * FROM orders ORDER BY order_date DESC;

ORDER_ID   CUSTOMER_I ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS
---------- ---------- ------------ ------------ --------------------
ORD010     CUS010     10-SEP-26          520.00 Shipped
ORD009     CUS009     09-SEP-26          200.00 Pending
ORD008     CUS008     08-SEP-26          360.00 Pending
ORD007     CUS007     07-SEP-26          420.00 Delivered
ORD006     CUS006     06-SEP-26          850.00 Shipped
ORD005     CUS005     05-SEP-26          180.00 Pending
ORD004     CUS004     04-SEP-26          400.00 Delivered
ORD003     CUS003     03-SEP-26          600.00 Shipped
ORD002     CUS002     02-SEP-26          250.00 Delivered
ORD001     CUS001     01-SEP-26          520.00 Delivered

10 rows selected.
```

4. Retrieving Unique Values Using DISTINCT:

1. Display unique product categories:

```
SQL> SELECT DISTINCT c.category_name FROM category c JOIN product p ON c.category_id = p.category_id;

CATEGORY_NAME
----------------------
Sustainable Fashion
Organic Food
Bamboo Products
Kitchen Items
Personal Care
Eco Stationery
Cloth Bags
Farming Products

8 rows selected.
```

2. Display different payment methods:

```
SQL> SELECT DISTINCT payment_mode FROM payment;

PAYMENT_MODE
---------------
Net Banking
Card
Cash
UPI
```

5. Searching Products Based on Conditions:

1. Find products priced between ₹20,000 and ₹40,000:

```
SQL> SELECT * FROM product WHERE price BETWEEN 20000 AND 40000;

no rows selected
```

2. Display products belonging to a particular category:

```
SQL> SELECT p.product_id, p.product_name, p.price, c.category_name FROM product p JOIN category c ON p.category_id = c.category_id WHERE c.category_name = '
Organic Food';

PRODUCT_ID PRODUCT_NAME                PRICE CATEGORY_NAME
---------- ------------------------- ------------ ----------------------
PRO001     Organic Oats                300.00 Organic Food
PRO002     Natural Honey               320.00 Organic Food
```

3. Find products available in inventory:

```
SQL> SELECT p.product_id, p.product_name, i.stock_quantity, i.stock_status FROM product p JOIN inventory i ON p.product_id = i.product_id WHERE i.stock_stat
us = 'Available' AND i.stock_quantity > 0;

PRODUCT_ID PRODUCT_NAME             STOCK_QUANTITY STOCK_STATUS
---------- ------------------------- -------------- ------------
PRO001     Organic Oats                        45 Available
PRO002     Natural Honey                       35 Available
PRO003     Aloe Face Cream                     40 Available
PRO004     Bamboo Comb                         90 Available
PRO005     Bamboo Tray                         30 Available
PRO006     Steel Lunch Box                     55 Available
PRO007     Eco Diary                           70 Available
PRO008     Cotton Tote Bag                     65 Available
PRO009     Eco Cotton Kurta                    25 Available
PRO010     Organic Seed Pack                   95 Available

10 rows selected.
```

4. Search products using product name:

```
SQL> SELECT * FROM product WHERE LOWER(product_name) LIKE '%honey%';

PRODUCT_ID PRODUCT_NAME              CATEGOR        PRICE STOCK_QUANTITY
---------- ------------------------- ------- ------------ --------------
PRO002     Natural Honey             CAT001        320.00             35
```

5. Find low-stock products:

```
SQL> SELECT product_id, product_name, stock_quantity FROM product WHERE stock_quantity <= 10;

no rows selected
```

6. Retrieving Customer and Product Information:

1. Display customer details along with their orders:

```
SQL> SELECT c.customer_id, c.customer_name, c.email, o.order_id, o.order_date, o.total_amount, o.order_status FROM customer c JOIN orders o ON c.customer_id
 = o.customer_id;

CUSTOMER_I CUSTOMER_NAME        EMAIL                       ORDER_ID   ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS
---------- -------------------- --------------------------- ---------- ------------ ------------ --------------------
CUS001     Sanjay Kumar         sanjay@gmail.com            ORD001     01-SEP-26          520.00 Delivered
CUS002     Harini Devi          harini@gmail.com            ORD002     02-SEP-26          250.00 Delivered
CUS003     Praveen Raj          praveen@gmail.com           ORD003     03-SEP-26          600.00 Shipped
CUS004     Keerthana S          keerthana@gmail.com         ORD004     04-SEP-26          400.00 Delivered
CUS006     Pavithra M           pavithra@gmail.com          ORD006     06-SEP-26          850.00 Shipped
CUS007     Ashwin Anand         ashwin@gmail.com            ORD007     07-SEP-26          420.00 Delivered
CUS010     Lavanya M            lavanya@gmail.com           ORD010     10-SEP-26          520.00 Shipped
CUS009     Surya Prakash        surya@gmail.com             ORD009     09-SEP-26          200.00 Pending
CUS008     Nithya R             nithya@gmail.com            ORD008     08-SEP-26          360.00 Pending
CUS005     Dinesh Kumar         dinesh@gmail.com            ORD005     05-SEP-26          180.00 Pending

10 rows selected.
```

2. Display product details with category information:

```
SQL> SELECT p.product_id, p.product_name, p.price, p.stock_quantity, c.category_name FROM product p JOIN category c ON p.category_id = c.category_id;

PRODUCT_ID PRODUCT_NAME                 PRICE STOCK_QUANTITY CATEGORY_NAME
---------- ------------------------ ---------- -------------- --------------------
PRO002     Natural Honey                320.00             35 Organic Food
PRO001     Organic Oats                 300.00             45 Organic Food
PRO003     Aloe Face Cream              250.00             40 Personal Care
PRO005     Bamboo Tray                  400.00             30 Bamboo Products
PRO004     Bamboo Comb                  100.00             90 Bamboo Products
PRO006     Steel Lunch Box              600.00             55 Kitchen Items
PRO007     Eco Diary                    140.00             70 Eco Stationery
PRO008     Cotton Tote Bag              180.00             65 Cloth Bags
PRO009     Eco Cotton Kurta             850.00             25 Sustainable Fashion
PRO010     Organic Seed Pack            110.00             95 Farming Products

10 rows selected.
```

3. Find customers who purchased a specific product:

```
SQL> SELECT DISTINCT c.customer_id, c.customer_name, p.product_name FROM customer c JOIN orders o ON c.customer_id = o.customer_id JOIN order_details od ON
o.order_id = od.order_id JOIN product p ON od.product_id = p.product_id WHERE p.product_name = 'Natural Honey';

CUSTOMER_I CUSTOMER_NAME        PRODUCT_NAME
---------- -------------------- ------------------------
CUS001     Sanjay Kumar         Natural Honey
CUS010     Lavanya M            Natural Honey
```

4. Display products purchased by each customer:

```
SQL> SELECT c.customer_name, o.order_id, p.product_name, od.quantity, od.price FROM customer c JOIN orders o ON c.customer_id = o.customer_id JOIN order_det
ails od ON o.order_id = od.order_id JOIN product p ON od.product_id = p.product_id;

CUSTOMER_NAME        ORDER_ID   PRODUCT_NAME              QUANTITY        PRICE
-------------------- ---------- ------------------------ -------- ------------
Sanjay Kumar         ORD001     Organic Oats                    1       200.00
Lavanya M            ORD010     Natural Honey                   1       320.00
Sanjay Kumar         ORD001     Natural Honey                   1       320.00
Harini Devi          ORD002     Aloe Face Cream                 1       250.00
Keerthana S          ORD004     Bamboo Tray                     1       400.00
Praveen Raj          ORD003     Steel Lunch Box                 1       600.00
Ashwin Anand         ORD007     Eco Diary                       3       140.00
Nithya R             ORD008     Cotton Tote Bag                 2       180.00
Dinesh Kumar         ORD005     Cotton Tote Bag                 1       180.00
Pavithra M           ORD006     Eco Cotton Kurta                1       850.00

10 rows selected.
```

7. Applying Multiple Filtering Conditions:

1. Find electronics products costing more than ₹600:

```
SQL> SELECT p.product_name, p.price, c.category_name FROM product p JOIN category c ON p.category_id = c.category_id WHERE p.price > 600 ;

PRODUCT_NAME                 PRICE CATEGORY_NAME
------------------------ ---------- --------------------
Eco Cotton Kurta             850.00 Sustainable Fashion
```

2. Search products containing the word “Cream”:

```
SQL> SELECT * FROM product WHERE LOWER(product_name) LIKE '%cream%';

PRODUCT_ID PRODUCT_NAME               CATEGOR        PRICE STOCK_QUANTITY
---------- ------------------------- ------- ------------ --------------
PRO003     Aloe Face Cream           CAT002         250.00             40
```

3. Find orders placed within a specific date range:

```
SQL> SELECT * FROM orders WHERE order_date BETWEEN DATE '2026-09-01' AND DATE '2026-09-10' ORDER BY order_date;

ORDER_ID   CUSTOMER_I ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS
---------- ---------- ------------ ------------ --------------------
ORD001     CUS001     01-SEP-26          520.00 Delivered
ORD002     CUS002     02-SEP-26          250.00 Delivered
ORD003     CUS003     03-SEP-26          600.00 Shipped
ORD004     CUS004     04-SEP-26          400.00 Delivered
ORD005     CUS005     05-SEP-26          180.00 Pending
ORD006     CUS006     06-SEP-26          850.00 Shipped
ORD007     CUS007     07-SEP-26          420.00 Delivered
ORD008     CUS008     08-SEP-26          360.00 Pending
ORD009     CUS009     09-SEP-26          200.00 Pending
ORD010     CUS010     10-SEP-26          520.00 Shipped

10 rows selected.
```

8.Basic Business Reports:

Report 1: Product Availability Report

This report displays product names, prices, stock quantities, and availability status.

```
SQL> SELECT p.product_name, p.price, i.stock_quantity, i.stock_status FROM product p LEFT JOIN inventory i ON p.product_id = i.product_id;

PRODUCT_NAME                PRICE STOCK_QUANTITY STOCK_STATUS
-------------------------- ------------ -------------- ------------
Organic Oats                300.00             45 Available
Natural Honey               320.00             35 Available
Aloe Face Cream             250.00             40 Available
Bamboo Comb                 100.00             90 Available
Bamboo Tray                 400.00             30 Available
Steel Lunch Box             600.00             55 Available
Eco Diary                   140.00             70 Available
Cotton Tote Bag             180.00             65 Available
Eco Cotton Kurta            850.00             25 Available
Organic Seed Pack           110.00             95 Available

10 rows selected.
```

Report 2: Customer Report

1.Find the total number of customers:

```
SQL> SELECT COUNT(*) AS total_customers FROM customer;

TOTAL_CUSTOMERS
---------------
             10
```

2. Display new customer registrations:

```
SQL> SELECT customer_id, customer_name, registration_date FROM customer WHERE registration_date >= DATE '2026-09-01' ORDER BY registration_date DESC;

no rows selected
```

Report 3: Order Report

1.Find the total number of orders:

```
SQL> SELECT COUNT(*) AS total_orders FROM orders;

TOTAL_ORDERS
-------------
           10
```

2. Find completed orders:

```
SQL> SELECT COUNT(*) AS completed_orders FROM orders WHERE order_status = 'Delivered';

COMPLETED_ORDERS
----------------
               4
```

3. Find pending orders:

```
SQL> SELECT COUNT(*) AS pending_orders FROM orders WHERE order_status IN('Pending','Shipped');

PENDING_ORDERS
--------------
             6
```

4. Find cancelled orders:

```
SQL> SELECT COUNT(*) AS cancelled_orders FROM orders WHERE LOWER(order_status) = 'cancelled';

CANCELLED_ORDERS
----------------
               0
```

Report 4: Product Performance Report

1. Find the highest-priced products:

```
SQL> SELECT product_id, product_name, price FROM product WHERE price = (SELECT MAX(price) FROM product);

PRODUCT_ID PRODUCT_NAME                    PRICE
---------- ------------------------- ------------
PRO009     Eco Cotton Kurta               850.00
```

2. Find the most-reviewed product:

```
SQL> SELECT p.product_name, COUNT(r.review_id) AS review_count FROM product p JOIN review r ON p.product_id = r.product_id GROUP BY p.product_id,
name HAVING COUNT(r.review_id) = (SELECT MAX(review_count) FROM (SELECT COUNT(*) AS review_count FROM review GROUP BY product_id));

PRODUCT_NAME             REVIEW_COUNT
------------------------ ------------
Cotton Tote Bag                     1
Bamboo Tray                         1
Aloe Face Cream                     1
Steel Lunch Box                     1
Eco Cotton Kurta                    1
Organic Oats                        1
Natural Honey                       1
Bamboo Comb                         1
Eco Diary                           1
Organic Seed Pack                   1
```

3. Display available products:

```
SQL> SELECT p.product_id, p.product_name, p.price, i.stock_quantity FROM product p JOIN inventory i ON p.product_id = i.product_id WHERE i.stock_status = 'A
vailable' AND i.stock_quantity > 0;

PRODUCT_ID PRODUCT_NAME                PRICE STOCK_QUANTITY
---------- ------------------------- ------------ --------------
PRO001     Organic Oats                  300.00             45
PRO002     Natural Honey                 320.00             35
PRO003     Aloe Face Cream               250.00             40
PRO004     Bamboo Comb                   100.00             90
PRO005     Bamboo Tray                   400.00             30
PRO006     Steel Lunch Box               600.00             55
PRO007     Eco Diary                     140.00             70
PRO008     Cotton Tote Bag               180.00             65
PRO009     Eco Cotton Kurta              850.00             25
PRO010     Organic Seed Pack             110.00             95

10 rows selected.
```